

EduWand Platform, Development and Delivery

Enrolment Growth Engine and AI Module

1. Executive Summary

Fovea Infotech proposes to build and deliver two modules of the EduWand platform together, in an eight week build: the Enrolment Growth Engine and the AI Module. The Enrolment Growth Engine captures every prospective-student enquiry and automates admissions follow up. The AI Module, made up of Lesson Studio and Assignment Lab, gives teachers AI-assisted lesson planning, personalised assignments, and grading.

The platform is delivered as two applications, a Unified App for Teacher, Student, and Parent roles, and a separate Admin and Leadership Dashboard, both built against one shared backend. Infrastructure runs on Amazon Web Services, Mumbai region, and the AI Module runs on Amazon Bedrock, using Claude Sonnet, Claude Haiku, and Nova Lite.

This proposal excludes commercial terms for Fovea Infotech's own services. Where AWS infrastructure cost is referenced, it is quoted from LessonForge's existing AWS Statement of Work, not estimated by Fovea Infotech.

2. Solution Architecture

Two applications share one backend and one data model.

- Unified App, mobile and web, for Teacher, Student, and Parent, role based views on one codebase
- Admin and Leadership Dashboard, web first, for School Administrators and Trust Leadership

The backend itself is one application. It is deployed across two production instances plus one worker instance for reliability and background processing, with a separate UAT copy for testing, as set out in Section 5. Both applications call this same backend.

3. Module 1, Enrolment Growth Engine

Captures every prospective-student enquiry from any channel as a structured record, then runs automated multi-channel follow up to convert enquiries into admissions. Built to run standalone, alongside a school's existing ERP, with no dependency on any other EduWand module.

3.1 Full Feature List

Enquiry Capture

- Omni-channel enquiry capture: phone call, walk-in, website form, referral, expo or event, social media
- Structured enquiry record: contact details, source, grade or board interest, status, owner, timestamps, notes
- Public website enquiry form, embeddable on the school's own site

- Bulk upload for enquiries collected at events or expos

Admissions Pipeline

- Configurable pipeline with stages: New, Contacted, Visit, Application, Admitted, Enrolled, or Lost with a reason
- Stage transitions tracked with timestamps and the staff member responsible
- Task assignment and reminders for counsellors and front desk staff
- Duplicate enquiry detection and merge, so the same family is not logged twice

Automated Follow Up

- Templated SMS and email reminders and notices
- Scheduled drip sequences for nurturing a lead over time
- Automatic response sent on receipt of a new enquiry
- Messaging consent capture and unsubscribe handling, per DPDP requirements

Admissions Workflow

- Application intake and document collection
- Admission confirmation step
- On confirmation, the system creates a student stub record: name, date of birth, class or section applied for, board, guardian name and contact, admission date, fee status placeholder, and the originating enquiry ID

Data Handoff to Incumbent ERP

- CSV export of admitted students back to the school's existing ERP
- User-definable column mapping, saved as a reusable export profile per school
- Export runnable on demand or on a schedule, with a full export log

Analytics

- Enquiry volume by source
- Conversion funnel and conversion rate
- Year on year enrolment growth tracking
- Counsellor performance view

3.2 Acceptance Criteria

- An enquiry can be logged in under approximately 30 seconds with all mandatory fields
- Automated follow up messages fire on schedule and are logged against the enquiry record
- Funnel and conversion analytics update in near real time
- Duplicate enquiries are detected at the point of entry
- Messaging consent is captured before any automated message is sent to a parent or guardian

4. Module 2, AI Module, Lesson Studio and Assignment Lab

Lets teachers use AI without prompting or switching between apps. Two sub-tools work together: Lesson Studio, for creating lesson plans, learning material, and research reports, and Assignment Lab, for creating and grading personalised assignments with live analytics for the teacher.

4.1 Full Feature List

Lesson Studio

- Lesson plan generation, for example in CBSE-mandated 5E format
- Content and research report generation from a topic or curriculum reference
- Board aware generation, adapting to CBSE, ICSE, or State board curriculum
- Outputs auto-saved to the classroom database, with no separate upload step

Assignment Lab

- Personalised assignment generation, suggesting a difficulty mix per student based on recent performance
 - Teacher-initiated design: the teacher reviews and approves the suggested mix before it is applied, rather than the system acting on a student automatically
 - Per-student opt out available, and recorded
- AI grading of submissions, with feedback and suggested next steps for each student
- Teacher override available on every AI-assigned grade
- Flags submissions that may need closer attention

Analytics

- Live per-student performance view, always available to the teacher
- Live per-class performance view
- Basic AI usage analytics surfaced to the Admin Dashboard: generations per teacher, grading turnaround

4.2 AI Models in Use

The AI Module runs on Amazon Bedrock. Tasks are routed to different models based on the nature of the task.

Model	Proposed Use
Claude Sonnet	Lesson plan generation and research report generation, where output quality and depth matter most
Claude Haiku	AI grading and feedback, and the personalisation suggestion step, where speed matters and tasks are more structured
Nova Lite	Lightweight tasks, short classification or tagging style operations, and any secondary or overflow load

4.3 Acceptance Criteria

- A lesson plan is generated in under approximately 60 seconds
- Personalisation is applied only after teacher approval, to the agreed share of assignment items

- AI grading is validated against a teacher-graded sample within an agreed accuracy tolerance
- Analytics update in near real time after a submission is graded
- All generated content is stored and retrievable for the full academic year

4.4 Data Safeguard

Personalising assignments from a student's performance history is treated with particular care in this build. Automated behavioural profiling of a child raises a data privacy question under India's DPDP Act. This module ships with a safer design:

- Personalisation is teacher-initiated and teacher-visible, not an automated profile acting on a child without review
- No persistent behavioural profile is created or acted on without a teacher's active decision at the point of use
- Per-student opt out is available and recorded

5. Infrastructure

Infrastructure runs on Amazon Web Services, Mumbai region, as defined in LessonForge's existing AWS Statement of Work.

5.1 Environment

Item	Detail
Cloud provider	Amazon Web Services
Region	Asia Pacific, Mumbai, ap-south-1
Environments	Production and UAT, logically separated within one VPC
Network	Public and private subnets across multiple Availability Zones

5.2 Compute

Server	Role	Instance Type
LF-APP-01 and LF-APP-02	Two instances of the same backend application, behind a load balancer, for redundancy	t4g.medium
LF-WORKER-01	Background and asynchronous job processing, same backend codebase, for example AI grading jobs and follow up messaging	t4g.large
LF-APP-UAT	UAT copy of the same backend application, isolated from production	t4g.medium
Bastion Server	Secure administrative access to private subnet resources	t3a.small

5.3 Database

- Amazon RDS for PostgreSQL, production, Multi-AZ, for automatic failover
- Amazon RDS for PostgreSQL, UAT, single-AZ, isolated from production data

5.4 Security and Delivery

- Amazon Route 53 for DNS
- AWS Web Application Firewall, protecting against common web exploits
- Application Load Balancer, distributing traffic across production app servers
- NAT Gateway, for secure outbound access from private subnets
- Amazon CloudWatch, for logs, metrics, and alerting

6. Technology Stack

Layer	Technology
Mobile	React Native, single codebase for iOS and Android
Web	React, Next.js
Backend	Node.js, NestJS
Database	Amazon RDS for PostgreSQL
Hosting	Amazon Web Services, Mumbai region
AI provider	Amazon Bedrock, Claude Sonnet, Claude Haiku, Nova Lite
Messaging	SMS gateway, DLT registered, transactional email
Authentication	OAuth 2.0, managed identity provider
File and object storage	Amazon S3
Monitoring	Amazon CloudWatch
Security	AWS WAF, Application Load Balancer, Bastion host, private subnets
CI / CD	GitHub Actions

7. Development Timeline, Eight Weeks

The Enrolment Growth Engine and the AI Module are built as two parallel tracks over eight weeks. A weekly review is held every Saturday for the duration of the build.

Week	Enrolment Growth Engine Track	AI Module Track
Week 1	Data model design, enquiry, pipeline, guardian entities. UI wireframes for Unified App and Dashboard. Environment setup.	Bedrock integration design. Prompt and output schema design for Lesson Studio and Assignment Lab. AI usage telemetry design.
Week 2	Authentication and roles. Enquiry capture, all channels. Pipeline and stage transitions.	Bedrock integration built. Lesson plan generation flow, first working version, using Claude Sonnet.
Week 3	Automated follow up engine, templating and scheduling. Duplicate detection. Task assignment and reminders.	Research report generation. Content auto-save to classroom database.
Week 4	Admissions workflow through to student stub record. CSV export with mapping profile.	Assignment Lab, teacher-initiated personalisation suggestion flow, using Claude Haiku. Per-student, per-class analytics view.
Week 5	Funnel and conversion analytics. Admin Dashboard, enrolment view. Consent capture, DPDP.	AI grading pipeline, first working version, with teacher override, using Claude Haiku.
Week 6	SMS integration, subject to DLT approval landing. Notification infrastructure hardening.	AI grading accuracy validation against a teacher-graded sample. AI usage analytics in Admin Dashboard.
Week 7	Functional and regression testing. Security pass, encryption, access control. Bug fixing.	Functional and regression testing. Prompt and output quality review across all three models. Bug fixing.
Week 8	User acceptance testing with LessonForge. Release notes and rollback plan. Go live readiness review.	User acceptance testing with LessonForge, Teacher role walkthrough. Fixes from UAT feedback.

8. Risks

Risk	Detail	Mitigation
AI running cost	Bedrock inference cost, per the AWS Statement of Work, is a material ongoing cost, driven by usage volume	Monitor actual spend from Week 5 onward once AI features are live in staging, against real teacher usage
Model behaviour differs across Claude Sonnet, Claude Haiku, and Nova Lite	Output quality and tone may vary by model, since tasks are routed across three different models	Validate each model's output against the acceptance criteria defined per feature, during Week 6 and Week 7 testing
Two vendors involved, CloudStok for AWS infrastructure, Fovea Infotech for application build	Coordination required at handoff points, environment access, deployment credentials	Joint kickoff between CloudStok and Fovea Infotech in Week 1, per the RACI defined in the AWS Statement of Work
Same team covers both tracks across eight weeks	Less slack per track than a single-module build would have	Fixed weekly split, engineers assigned specific days per track, QA begins Week 3

9. Next Steps

- Confirm AWS account access and Bedrock model access for the Fovea Infotech team
- Confirm the RACI handoff points between CloudStok and Fovea Infotech for Week 1
- Initiate DLT registration for SMS immediately
- Schedule kickoff and Week 1 discovery session

End of proposal. Fovea Infotech.